

Hazard Analysis Software Engineering

Team 11, technically functional

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
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[You are free to modify this template. —SS]

1 Introduction

The team’s definition of a hazard is an “unsafe” condition that may lead to an unintended event that causes an undesirable outcome [Ippolito and Wallace, 1995]. In relation to software development, hazards can be understood as the underlying design decisions that result in potential harm. These risks become apparent when the hazards are not addressed prior to development. These hazards can include technical failures, privacy/security, usability and accessibility risks.

The project’s goal is to develop a tool that provides guidance for a physiotherapist’s prescribed home exercises. The user’s form is analyzed and prompts for adjustments are given. While this can increase the rate and efficacy of recovery, it introduces several hazards. [See the Section 3 in the Problem Statement document for more on project goals.](#) The purpose of this document is to identify those hazards, outline assumptions and propose solutions to limit their presence and their frequency.

2 Scope and Purpose of Hazard Analysis

This document outlines the hazards that may occur while operating the tool. The key areas for scope include:

- Technical Hazards: Inability to process user footage, incorrect identification of exercise from different planes, or inaccurate data comparisons, which can result in performance issues and impact the functionality of the tool. Furthermore, these hazards can result in misleading feedback, which could contribute to improper and unsafe exercise techniques.
- Usability Hazards: Usability ensures that the tool has the potential for successfully being adopted. All of the inputs to the system, such as the recording of the exercise, are generated from the user. A user-friendly interface guarantees ease of use, which encourages utilization of this tool. On the other hand, incorrect, on-screen prompts may cause confusion while the user performs the exercise. This may reduce usability and lead to ambiguous guidelines.
- Privacy/Security Hazards: Unnecessary data collection and storage increases the potential of unauthorized access and misuse of personal information. Such situations, such as data breaches, may lead to catastrophic outcomes, which will result in lower adoption rate.

The purpose of this document is to identify risks and their associated impact to determine ways to mitigate them and identify any future occurrences

3 System Boundaries and Components

To divide the system boundaries we outline the key functions of the application.

- Authentication module
 - The user will log in to the application to access prior user data.
- Video streaming module
 - The user will video stream themselves performing a physio exercise pertaining to the knee joint.
- Analysis module
 - The input of the user performing an exercise will be assessed against the requirements and guidelines defined in the backend.
- Reporting module
 - The system will generate analytics from the analysis module which packages the information to be viewed by the user or physiotherapist.

In addition we will consider the following within the hazard analysis:

- A device with a camera
 - To allow users to record themselves during the motion and to receive feedback on the exercise performed.
- A database where user data will be stored

These system boundaries outline what is covered within this hazard analysis.

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

References

Laura M. Ippolito and Dolores R. Wallace. *A Study on Hazard Analysis in High Integrity Software Standards and Guidelines*. 1995. URL <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir5589.pdf>. U.S. Department of Commerce, Technology Administration, National Institute of Standards and Technology.